RESEARCH ARTICLE

Digital Westphalia: AI Sovereignty as the New Frontier of Geopolitical Competition in 2026

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ABSTRACT

This paper introduces the concept of 'Digital Westphalia' as a theoretical framework for understanding the emerging tripartite competition between the United States, China, and the European Union over artificial intelligence sovereignty. Drawing on classical sovereignty theory and adapting it for the digital age, the study proposes an original 'AI Sovereignty Index' comprising six dimensions: compute infrastructure, foundation model autonomy, data governance, talent pipeline retention, regulatory independence, and military AI integration. Through comparative analysis of national AI strategies, investment patterns, and regulatory architectures across 2020-2026, this research demonstrates that AI governance has become the defining axis of great power competition, superseding traditional domains of territorial and economic sovereignty. The paper argues that the international system is entering a period of 'Digital Westphalianism' characterized by the assertion of sovereign control over AI ecosystems, the fragmentation of the global digital commons, and the emergence of competing normative orders for artificial intelligence governance. The findings reveal significant asymmetries in AI sovereignty capabilities that carry profound implications for international security, alliance structures, and the future of strategic stability.

Keywords: AI Sovereignty, Digital Geopolitics, Great Power Competition, AI Governance, Digital Westphalia, Strategic Technology, Compute Infrastructure, Foundation Models

1. Introduction

The international order is being fundamentally reshaped by the competition for artificial intelligence supremacy. As governments increasingly treat AI assets—foundation models, training data, and computing infrastructure—as national security priorities and critical infrastructure, the traditional Westphalian conception of sovereignty is being extended into the digital domain. This paper introduces the concept of 'Digital Westphalia' to describe the emerging international order in which states assert sovereign control over their AI ecosystems with the same vigor historically reserved for territorial boundaries and economic systems.

The year 2026 represents a critical inflection point in this transformation. The United States has channeled unprecedented government investment into domestic AI infrastructure through executive orders and CHIPS Act Phase II allocations. China's 15th Five-Year Plan (2026–2030) has designated technology self-reliance as a core strategic priority. The European Union has proposed the AI Continent Action Plan and the Cloud and AI Development Act, seeking to establish a distinctive 'third way' between American market-driven and Chinese state-directed approaches.

This paper makes three original contributions to the literature. First, it proposes a comprehensive theoretical framework—Digital Westphalia—that synthesizes sovereignty theory with the unique characteristics of AI as a strategic technology. Second, it develops a novel 'AI Sovereignty Index' comprising six measurable dimensions that enable systematic cross-national comparison. Third, it provides empirical analysis of how the tripartite competition between the US, China, and the EU is restructuring the international system, with particular attention to the implications for middle powers, alliance structures, and strategic stability.

2. Theoretical Framework: From Westphalia to Digital Westphalia

The Peace of Westphalia (1648) established the foundational principle of state sovereignty—that each state possesses supreme authority within its territorial boundaries. This paper argues that we are witnessing a parallel moment of systemic transformation in which sovereignty is being redefined to encompass the digital domain, particularly artificial intelligence.

2.1 The Sovereignty-Technology Nexus

The relationship between sovereignty and technology is not new. The Industrial Revolution created new dimensions of national power that reshaped the balance of power in the 19th century. Nuclear technology in the 20th century introduced the concept of 'technological sovereignty' in the realm of strategic deterrence. AI represents the third major inflection point in this trajectory, distinguished by several unique characteristics that amplify its sovereign significance.

Unlike nuclear technology, which required massive state investment and remained largely within government control, AI development exists at the intersection of public and private power. The concentration of frontier AI capabilities among a handful of corporations—predominantly American—creates a novel form of power asymmetry that challenges traditional state-centric conceptions of sovereignty. Governments must now negotiate not only with other states but with corporate entities that possess capabilities rivaling or exceeding those of many nation-states.

2.2 Defining AI Sovereignty

This paper defines AI sovereignty as a state's capacity to exercise effective control over the development, deployment, governance, and strategic application of artificial intelligence within its jurisdiction, while maintaining meaningful autonomy in these domains relative to external actors. This definition encompasses six measurable dimensions, which together constitute the AI Sovereignty Index proposed in this study.

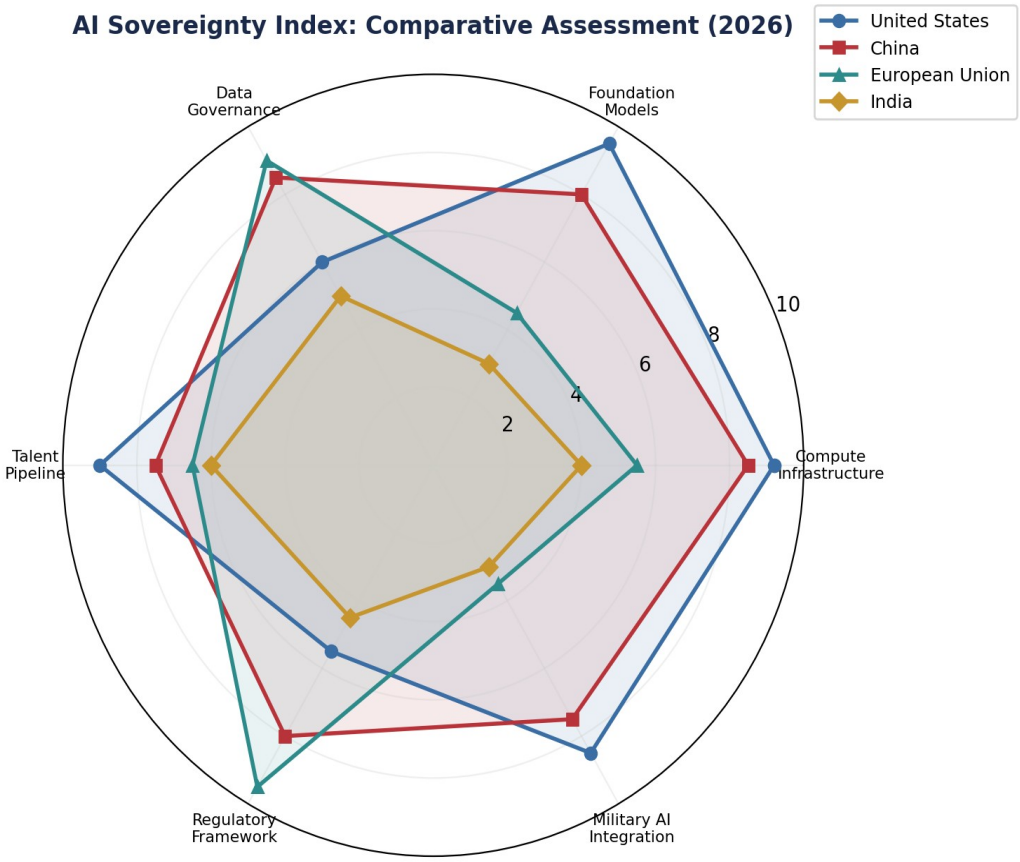

Figure 1: AI Sovereignty Index — Comparative Assessment of Major Powers (2026)

3. The AI Sovereignty Index: A Novel Analytical Framework

The AI Sovereignty Index (AISI) proposed in this paper represents an original contribution to the measurement of state capacity in the digital age. Drawing on methodologies from the OECD Government AI Readiness Index, the Stanford AI Index, and the Global Cybersecurity Index, the AISI integrates six dimensions into a comprehensive assessment framework.

3.1 Compute Infrastructure (Weight: 20%)

Compute infrastructure encompasses the hardware ecosystem required for AI development, including advanced semiconductor fabrication, GPU/TPU manufacturing, and data center capacity. The United States maintains a commanding lead through NVIDIA's dominance in AI accelerators and the concentration of hyperscale data centers on American soil. China has made

substantial progress through Huawei's Ascend chips and massive state-funded data center construction, though the US semiconductor export controls imposed since 2022 have constrained access to the most advanced chips. The EU significantly lags in this dimension, possessing no indigenous frontier chip manufacturer.

3.2 Foundation Model Autonomy (Weight: 20%)

This dimension measures a state's capacity to develop, train, and deploy frontier AI models independently. The United States leads overwhelmingly, hosting the organizations behind GPT-5, Claude 4, and Gemini Ultra. China has developed competitive domestic models including DeepSeek, Qwen, and Ernie, though performance gaps persist at the frontier. The EU has no indigenous frontier model of comparable capability, creating a dependency relationship that fundamentally constrains its AI sovereignty.

3.3 Data Governance (Weight: 15%)

Data governance captures the regulatory and institutional frameworks governing data collection, storage, processing, and cross-border transfer. The EU leads decisively through the GDPR framework, the Data Act, and the AI Act's data governance provisions. China's cybersecurity and data security regime provides comprehensive state control over data flows. The United States, lacking comprehensive federal data privacy legislation, scores lowest among the three powers despite the volume of data generated within its borders.

3.4 Talent Pipeline (Weight: 15%)

The talent dimension assesses the capacity to produce, attract, and retain AI researchers and engineers. The United States benefits from the gravitational pull of Silicon Valley and leading research universities, though immigration policy volatility creates vulnerability. China has invested massively in STEM education and now produces more AI-related PhDs than any other country, but faces brain drain challenges. The EU possesses strong university systems but struggles with talent retention as researchers migrate to higher-compensation environments in the US and increasingly in the Gulf states.

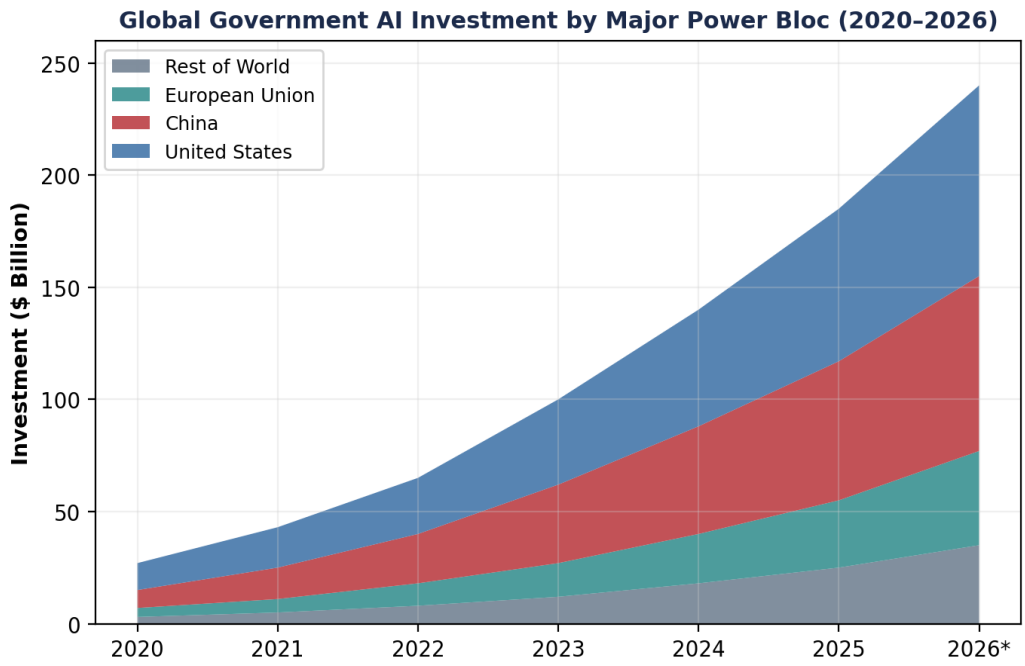

Figure 2: Global Government AI Investment by Major Power Bloc (2020–2026)

3.5 Regulatory Independence (Weight: 15%)

Regulatory independence measures the degree to which a state can set and enforce its own AI governance norms without subordination to external frameworks. The EU scores highest, having established the world's most comprehensive AI regulatory architecture through the AI Act. China's regulatory framework, while less transparent, provides the state with extensive control over AI deployment within its borders. The United States, under the current administration's deregulatory approach, has prioritized innovation over regulation, which paradoxically reduces its 'regulatory sovereignty' by ceding norm-setting power to corporate actors and foreign regulators.

3.6 Military AI Integration (Weight: 15%)

This dimension assesses the incorporation of AI into defense and intelligence capabilities. The United States leads through programs such as the Replicator initiative, JADC2, and extensive AI integration across intelligence agencies. China has made military AI a centerpiece of its military-civil fusion strategy. The EU, hampered by fragmented defense structures and limited defense-industrial AI investment, significantly trails in this dimension, though individual member states such as France and Estonia are making targeted advances.

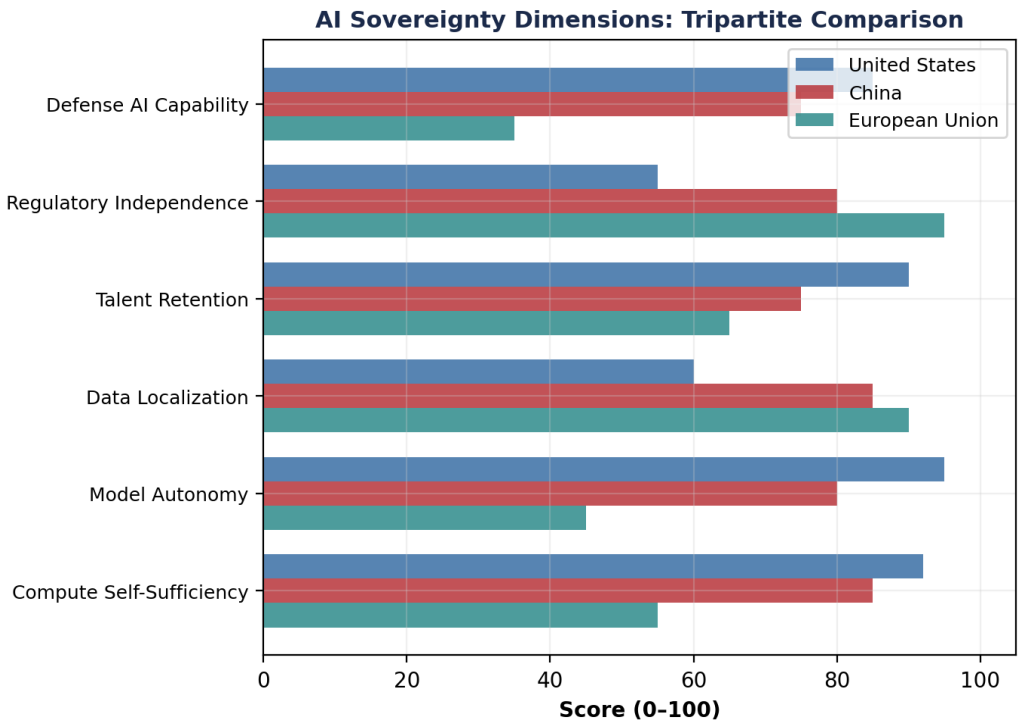

Figure 3: AI Sovereignty Dimensions — Tripartite Comparison Across Six Dimensions

4. The Tripartite Competition: Dynamics and Implications

4.1 The American Model: Market-Led Innovation with Strategic Guardrails

The United States' AI sovereignty strategy rests on the symbiotic relationship between the world's most innovative private sector and strategic government intervention through export controls, infrastructure investment, and selective regulation. The January 2026 Executive Order on AI Infrastructure committed significant federal resources to domestic compute capacity, while the CHIPS Act Phase II allocations directed billions toward semiconductor manufacturing onshoring. This approach maximizes innovation velocity but creates sovereign vulnerabilities through corporate concentration and the absence of comprehensive data governance.

4.2 The Chinese Model: State-Directed Technological Self-Reliance

China's AI sovereignty strategy is characterized by the comprehensive mobilization of state resources toward technological self-reliance. The 15th Five-Year Plan (2026–2030) designates AI as a strategic priority sector, building on the substantial foundations laid by the New Generation AI Development Plan. China's approach integrates industrial policy, military-civil fusion, and regulatory control into a coherent sovereignty framework. The export control escalation by the United States has paradoxically accelerated China's drive toward AI self-sufficiency, particularly in chip design and alternative computing architectures.

4.3 The European Model: Normative Power and Regulatory Sovereignty

The European Union's AI sovereignty strategy leverages its distinctive asset—regulatory power—to establish a 'third way' in the global AI competition. The AI Act, the AI Continent Action Plan, and the proposed Cloud and AI Development Act represent an ambitious effort to project European values and standards onto the global AI ecosystem. However, the EU's critical weakness lies in its lack of indigenous frontier AI capabilities, creating a fundamental tension between regulatory ambition and technological dependency.

5. Strategic Implications for International Security

5.1 Alliance Restructuring

The emergence of AI sovereignty as a primary axis of competition is driving a fundamental restructuring of alliance relationships. Traditional security alliances are being supplemented—and in some cases complicated—by AI-specific partnerships. The Australia-UK-US (AUKUS) arrangement has expanded to encompass AI cooperation. Japan and South Korea, despite historical tensions, are exploring AI semiconductor supply chain integration. India's 'multi-alignment' strategy in AI—engaging with the US, EU, and indigenous development simultaneously—represents a novel approach to navigating the digital sovereignty landscape.

5.2 The Middle Power Dilemma

Middle powers face an acute dilemma in the emerging Digital Westphalian order. States such as India, Brazil, Indonesia, and the Gulf nations must choose between aligning with one of the three major AI sovereignty blocs or pursuing costly indigenous AI development. This paper identifies a growing 'AI non-alignment' movement among Global South states seeking to preserve strategic autonomy by maintaining access to multiple AI ecosystems while avoiding exclusive dependency on any single power bloc.

5.3 Strategic Stability Risks

The fragmentation of the global AI ecosystem into sovereign blocs introduces new risks to strategic stability. Asymmetries in military AI capabilities could undermine deterrence frameworks premised on mutual capability awareness. The absence of international norms governing AI in military contexts—analogous to the nuclear non-proliferation regime—creates a regulatory vacuum in which strategic miscalculation becomes increasingly likely. The paper recommends the establishment of an 'AI Strategic Stability Dialogue' between the US and China, modeled on Cold War arms control frameworks but adapted for the unique characteristics of AI technology.

6. Comparative Analysis: AI Sovereignty Profiles

Dimension	United States	China	European Union
Compute Infrastructure	Dominant (NVIDIA, hyperscale)	Rising (Huawei Ascend)	Dependent (no frontier fab)
Foundation Models	Leading (GPT-5, Claude, Gemini)	Competitive (DeepSeek, Qwen)	Absent at frontier
Data Governance	Weak (no federal law)	Comprehensive (state control)	Strongest (GDPR, AI Act)
Talent Pipeline	Strong (immigration-dependent)	Growing (brain drain risk)	Moderate (retention issues)
Regulatory Independence	Low (corporate-led)	High (state-directed)	Highest (normative power)
Military AI	Advanced (Replicator, JADC2)	Advancing (MCF strategy)	Fragmented (nation-level)

Table 1: Comparative AI Sovereignty Profiles of the Three Major Powers

7. Conclusion

The international system is entering a period of Digital Westphalianism in which the assertion of sovereign control over artificial intelligence ecosystems has become a primary axis of great power competition. This paper's AI Sovereignty Index reveals significant asymmetries across the three major power blocs: the United States leads in innovation and military application but faces governance gaps; China pursues comprehensive state-directed self-reliance but confronts persistent frontier capability constraints; and the European Union projects regulatory sovereignty while lacking indigenous technological foundations.

The implications for international security are profound. The fragmentation of the global AI ecosystem threatens interoperability, increases the risk of strategic miscalculation, and creates new dependencies that reshape alliance structures. Middle powers face an acute sovereignty dilemma as they navigate between competing AI blocs. The absence of international governance frameworks for AI—particularly in the military domain—represents a dangerous vacuum that demands urgent diplomatic attention.

This paper calls for the establishment of a multilateral AI Strategic Stability Dialogue, the development of transparency and confidence-building measures for military AI, and the creation of interoperability standards that prevent the complete fragmentation of the global AI ecosystem. The choices made by states in 2026 and beyond will determine whether the Digital Westphalian order evolves into a framework for managed competition or descends into a technological cold war with consequences far exceeding those of the original.

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